

2023 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

Summary of Question:

A bike (85 kg) pulls a trailer (28 kg). Forward force 125 N, friction 45 N on bike, 15 N on trailer.

(a) Show that acceleration = 0.58 m s^{-2}

Total mass = $85 + 28 = 113 \text{ kg}$

Unbalanced force = $125 - 45 - 15 = 65 \text{ N}$

$a = F/m = 65 / 113 = 0.58 \text{ m s}^{-2}$

Answer: 0.58 m s^{-2}

(b) Magnitude of the tension

For trailer ($m = 28 \text{ kg}$): $T - 15 = 28 \times 0.58$

$T = 16.24 + 15 = 31.2 \text{ N}$

Answer: 31 N

(c) Does the tension change?

As speed increases, friction increases.

Since acceleration stays constant (0.58 m s^{-2}), the unbalanced force must remain the same.

Therefore, tension must increase to balance the greater friction.

Answer: Tension increases

Final Answers

(a) 0.58 m s^{-2}

(b) 31 N

(c) Increases

Revision Tips

- Always calculate unbalanced force before acceleration.
- For coupled systems, analyse one part to find tension.
- Keep signs consistent: define forward as positive.
- If a remains constant while resistive forces increase, the pulling/tension must also increase.